

Cross-Level Residual Frequency-Guided Multi-Scale Refinement with a ConvNeXt Encoder for Polyp Segmentation

Abstract

Accurate colorectal polyp segmentation remains challenging because lesions vary in scale, morphology, contrast, and boundary appearance. Conventional multi-scale convolution blocks aggregate different receptive fields without explicitly using sample-specific frequency information from deep semantic features. We present a ConvNeXt-Tiny U-shaped framework that builds on a bottleneck Frequency-Aware Feature Enhancement Module (FAFEM) and a Multi-Scale Convolution Block (MSCB), and introduce a Residual Frequency-Guided MSCB (RFG-MSCB). RFG-MSCB reuses the low- and high-frequency descriptor produced by FAFEM to assign sample-dependent weights to the MSCB multi-kernel branches at the Stage-3 encoder-decoder fusion. The guided response is introduced as a bounded residual deviation from ordinary MSCB aggregation, preserving the conventional multi-scale path as a reference. We evaluated the model on Kvasir-SEG, CVC-ClinicDB, CVC-300, CVC-ColonDB, and ETIS-LaribPolypDB over three random seeds. The model obtained mDice/mIoU of $0.9313 \pm 0.0044/0.8826 \pm 0.0062$ on Kvasir-SEG and $0.9370 \pm 0.0024/0.8877 \pm 0.0038$ on CVC-ClinicDB. On the three external datasets, mDice ranged from 0.7870 ± 0.0113 on ETIS-LaribPolypDB to 0.8821 ± 0.0016 on CVC-300. In the three-seed ablation, adding residual frequency-guided coupling to FAFEM + MSCB increased Kvasir-SEG mDice from 0.9304 ± 0.0022 to 0.9313 ± 0.0044 , an observed difference of +0.08 percentage points. These results support bounded cross-level frequency guidance as a viable refinement strategy, while performance degradation on the more challenging external datasets indicates remaining generalization limitations.

Introduction

Colorectal polyps are abnormal tissue growths that can develop into malignant lesions if they are not detected and removed at an early stage. Colonoscopy is widely used for the examination of the lower gastrointestinal tract and enables clinicians to identify and remove suspicious lesions. Nevertheless, visual inspection remains challenging because polyp appearance varies considerably across patients and imaging conditions. Variations in lesion size, morphology, texture, contrast, and location, together with specular reflections, motion blur, intestinal folds, bubbles, and other visual artifacts, can complicate accurate delineation of polyp regions. Automated polyp segmentation therefore remains an important problem in computer-aided colonoscopy, where reliable pixel-level localization may assist subsequent analysis and clinical decision support.

Accurate polyp segmentation requires a model to simultaneously capture high-level semantic information and preserve sufficiently detailed spatial representations. U-shaped encoder-decoder architectures are well suited to this task because they combine progressively abstract encoder features with higher-resolution representations through skip connections. U-Net established this general design, while subsequent architectures such as UNet++ further demonstrated the importance of multi-level feature interaction and hierarchical decoding [1, 2]. However, repeated downsampling reduces spatial resolution, and fixed decoder operations may not always provide sufficient flexibility for lesions that vary considerably in scale and visual frequency content.

Recent transformer-based and hybrid architectures have improved long-range dependency modeling in medical image segmentation. TransUNet combines convolutional representations with transformer encoding, while Polyp-PVT applies hierarchical transformer features and cross-level aggregation to polyp segmentation [3, 4]. These approaches demonstrate the importance of broad contextual modeling, but they can introduce additional architectural complexity. Modern convolutional networks such as ConvNeXt provide

an alternative by retaining the efficiency and spatial inductive bias of convolution while incorporating design principles that improve representation capacity. ConvNeXt employs large-kernel depthwise convolution, inverted channel expansion, Layer Normalization, GELU activation, layer scaling, and stochastic depth, making it a strong hierarchical encoder for dense prediction tasks [5].

Multi-scale representation is particularly important in polyp segmentation because lesions may occupy very small regions or extend over a large portion of the image. Multi-Scale Convolution Blocks (MSCBs) address this issue by combining convolutional branches with different receptive fields, allowing a network to aggregate local and broader spatial responses. However, conventional multi-branch aggregation generally combines the branch responses using a fixed formulation that is shared across all input images. Consequently, the relative contribution of different receptive fields is not explicitly conditioned on the frequency characteristics of each individual sample.

Frequency-aware feature processing provides another complementary perspective. Frequency-Aware Feature Enhancement Modules (FAFEMs) decompose deep representations into low- and high-frequency components and use these components to enhance feature responses. Low-frequency representations mainly preserve broader semantic structures, whereas high-frequency representations emphasize local spatial variations and fine structural information. Existing frequency-aware modules are typically used to improve the feature representation at the location where the frequency decomposition is performed. However, the frequency descriptor extracted from deep semantic representations can potentially provide useful global information for guiding later multi-scale processing.

This observation motivates the central idea of this work. Instead of treating bottleneck frequency enhancement and decoder multi-scale refinement as independent operations, we establish an explicit cross-level interaction between them. The frequency descriptor produced by a bottleneck FAFEM is propagated to the first decoder stage and used to regulate the relative contributions of the multi-scale branches in an MSCB. In this way, the receptive-field mixture of the decoder becomes sample dependent rather than fixed across all images.

Directly replacing the original MSCB aggregation with a fully frequency-guided formulation could, however, unnecessarily perturb a stable multi-scale representation. To address this issue, we introduce the bounded residual guidance strategy formalized in Eq. (23). Here, M_{ord} denotes the ordinary multi-scale aggregation, M_{guide} denotes the frequency-conditioned aggregation, and s is a bounded learnable guidance-strength parameter. This formulation preserves the conventional MSCB pathway as a reference and allows the model to progressively learn frequency-dependent deviations during optimization.

Based on this mechanism, we develop a ConvNeXt-Tiny U-shaped segmentation framework in which the deepest encoder feature is first enhanced using an existing FAFEM. The resulting low-/high-frequency descriptor is retained and transferred across network levels. After the first decoder feature is fused with the Stage-3 encoder representation, the proposed Residual Frequency-Guided Multi-Scale Convolution Block (RFG-MSCB) uses this descriptor to generate sample-adaptive weights for the multi-kernel MSCB branches. The remaining decoder stages use lightweight conventional fusion and depthwise-separable convolution, keeping the additional architectural modification localized to the first encoder-decoder fusion stage.

The principal contributions of this work are summarized as follows:

- We introduce a frequency-conditioned multi-scale refinement mechanism, termed RFG-MSCB, which reuses the frequency descriptor generated by an existing bottleneck FAFEM to adaptively weight the multi-kernel branches of an existing MSCB. The contribution lies in the coupling between the frequency descriptor and the multi-scale branches rather than in FAFEM or MSCB themselves.
- We formulate the frequency-guided aggregation as a bounded residual deviation from the conventional MSCB response. This design preserves the ordinary multi-scale pathway as a reference while allowing a controlled, learnable, sample-dependent modification of branch importance.
- We introduce cross-level bottleneck-to-decoder frequency guidance by transferring the bottleneck frequency descriptor to the Stage-3 decoder refinement stage. This connects deep semantic frequency information with higher-resolution multi-scale feature processing while leaving the remaining decoder

stages lightweight and unchanged.

- We evaluate the resulting ConvNeXt-Tiny segmentation architecture on multiple public colorectal polyp datasets and perform controlled ablation experiments to examine the contribution of frequency-aware enhancement, residual frequency guidance, and the selected Stage-3 placement.

Related work

Encoder–decoder architectures for medical image segmentation

Fully convolutional encoder–decoder networks remain a central design pattern for biomedical image segmentation. U-Net introduced a symmetric contracting and expanding architecture in which high-resolution encoder features are transferred to the decoder through skip connections, allowing semantic context and spatial detail to be combined during prediction [1]. UNet++ further showed that the structure of skip pathways is important by using nested dense connections to reduce the semantic discrepancy between encoder and decoder representations [2]. These works established the importance of hierarchical feature reuse, but they do not prescribe how the decoder should adapt its receptive-field mixture to the image-specific content of a polyp.

Polyp-specific convolutional models have explored efficient alternatives to large attention-based designs. HarDNet-MSEG combines a low-memory-traffic convolutional backbone with a lightweight decoder and demonstrates that strong polyp segmentation can be obtained without a transformer-heavy architecture [6]. PraNet introduced a parallel partial decoder and reverse attention to progressively refine uncertain polyp regions [7]. These methods highlight the value of efficient decoding and iterative refinement, but their main mechanisms are not designed to use frequency information from the bottleneck to regulate multi-scale convolutional branches at an earlier decoder stage.

Modern convolutional encoders provide another route to high-capacity dense prediction. ConvNeXt revisits convolutional network design with large-kernel depthwise convolution, inverted channel expansion, Layer Normalization, GELU activation, layer scaling, and stochastic depth, producing a hierarchical backbone with strong representation capacity [5]. In this work, ConvNeXt-Tiny is used as the encoder, while the novelty is concentrated in the decoder interaction between bottleneck frequency information and Stage-3 multi-scale convolution.

Transformer and contextual polyp segmentation

The large variation in polyp size, shape, color, and boundary contrast motivates architectures that capture both local appearance and broader anatomical context. Transformer-based approaches address this need by modeling long-range spatial relationships. TransUNet combines convolutional feature extraction with a transformer encoder for medical image segmentation [3], and Polyp-PVT applies pyramid vision transformer features and specialized fusion to polyp segmentation [4]. Recent polyp-segmentation studies have continued this direction through contrastive transformer modeling, contextual information flow, phase-wise feature pyramids, and pyramid-transformer feature fusion [8, 9, 10, 11].

These approaches show that global context and cross-level interaction are useful for polyp segmentation. However, attention-based or multi-stage contextual architectures can introduce additional architectural complexity. The present model therefore keeps a convolutional encoder–decoder form and asks a narrower question: whether a compact bottleneck frequency descriptor can guide the relative weighting of efficient multi-scale convolutional branches in the first decoder fusion stage.

Efficient multi-scale convolutional decoding

Multi-scale convolution is widely used because polyps may appear as small, subtle lesions or occupy a large image region. Parallel kernels and dilated or depthwise branches allow a decoder to aggregate local and broader responses without relying entirely on self-attention. EMCAD introduced an efficient multi-scale convolutional attention decoder for medical image segmentation and uses Multi-Scale Convolution Blocks

(MSCBs) as part of its lightweight decoding design [12]. In an MSCB, multiple convolutional branches provide different receptive fields and are combined to form a multi-scale response.

MSCB is prior work and is not introduced in this paper. The limitation relevant to the present method is that a conventional MSCB aggregation treats the branch mixture as fixed by the block design rather than conditioning the branch weights on the image-specific frequency characteristics extracted at the bottleneck. Our RFG-MSCB retains the ordinary MSCB path as a reference and adds only a bounded, frequency-conditioned residual deviation from that path.

Frequency-aware and adaptive feature recalibration

Attention and recalibration mechanisms provide a general precedent for using compact descriptors to control feature responses. Squeeze-and-Excitation Networks use global channel descriptors to recalibrate convolutional features [13], while Selective Kernel Networks adaptively weight convolutional branches with different receptive fields [14]. These methods are important novelty comparators because they show that descriptor-driven reweighting and adaptive receptive-field selection are established ideas.

Frequency-aware attention provides a closer conceptual link. FcaNet interprets channel attention through frequency analysis and uses frequency components for channel attention [15]. In polyp segmentation, Polyp-Mamba combines multi-frequency perception with a gated selection mechanism to model frequency-sensitive representations [16]. These studies motivate the usefulness of frequency information, but they do not correspond to the specific coupling used here: a frequency descriptor produced by a bottleneck FAFEM is transferred across levels and used to regulate Stage-3 MSCB branch weights.

The Frequency-Aware Feature Enhancement Module (FAFEM) is also treated as prior work in this paper. It is adopted at the bottleneck to enhance deep features and produce a low-/high-frequency descriptor. The contribution of this work is not FAFEM itself, but the reuse of its descriptor as cross-level guidance for multi-scale convolution in the decoder. The exact original FAFEM citation should be inserted once the source title, DOI, arXiv identifier, or publisher record is verified.

Boundary and recent polyp-segmentation refinements

Weak and visually ambiguous lesion boundaries remain a major challenge in polyp segmentation. Boundary-aware methods often introduce explicit edge supervision, auxiliary contour branches, image-pyramid high-frequency cues, or reverse attention. MEGANet employs multi-scale edge-guided attention for weak-boundary polyp segmentation [17], and MSBP-Net performs multi-scale boundary prediction to improve contour delineation [18]. Other recent methods model uncertainty, contextual flow, frequency-sensitive representations, and multi-stage refinement [19, 9, 16, 20].

The proposed architecture does not add an explicit boundary branch, edge label, or boundary loss. Boundary accuracy is instead expected to benefit indirectly from adaptive multi-scale decoding and from the use of frequency-guided branch weights at the first decoder fusion stage. Consequently, claims about boundary improvement should be supported by the experimental results rather than by describing a dedicated boundary module.

Positioning of the proposed method

Existing work has improved polyp segmentation through U-shaped skip reuse, reverse attention, transformer-based context modeling, efficient convolutional decoding, frequency-aware representation learning, and boundary-aware refinement. The proposed method is positioned at the intersection of efficient multi-scale convolution and frequency-aware recalibration.

Specifically, the final model uses a ConvNeXt-Tiny encoder, a bottleneck FAFEM, a lightweight residual bottleneck, Stage-3 SimpleFusion, the proposed Residual Frequency-Guided MSCB, and standard remaining decoder stages. FAFEM and MSCB are reused prior modules. The proposed contribution is the coupling between them: the bottleneck FAFEM frequency descriptor adaptively weights MSCB branches at Stage 3, the guided response is introduced as a bounded residual deviation from ordinary MSCB aggregation, and the

guidance forms a bottleneck-to-Stage-3 cross-level connection. This framing separates the new RFG-MSCB mechanism from the prior components on which it is built.

Method

Overall Architecture

The proposed framework follows a U-shaped encoder-decoder architecture built upon a ConvNeXt-Tiny encoder. The central objective is to exploit frequency information extracted from deep semantic features to adaptively regulate multi-scale feature processing during decoding. Rather than independently applying frequency enhancement and multi-scale convolution, the proposed model establishes a cross-level interaction between the bottleneck and the first decoder stage.

Let

$$I \in \mathbb{R}^{3 \times H \times W}$$

denote an input colonoscopy image. The ConvNeXt-Tiny encoder extracts four hierarchical feature maps,

$$\{F_1, F_2, F_3, F_4\} = E(I), \quad (1)$$

where

$$F_1 \in \mathbb{R}^{96 \times H/4 \times W/4}, \quad F_2 \in \mathbb{R}^{192 \times H/8 \times W/8},$$

$$F_3 \in \mathbb{R}^{384 \times H/16 \times W/16}, \quad F_4 \in \mathbb{R}^{768 \times H/32 \times W/32}.$$

For an input resolution of 352×352 , their spatial dimensions are 88×88 , 44×44 , 22×22 , and 11×11 , respectively.

The deepest encoder feature F_4 is first processed by a Frequency-Aware Feature Enhancement Module (FAFEM), which produces an enhanced bottleneck representation together with a compact frequency descriptor. The enhanced representation is passed through a lightweight residual bottleneck, whereas the frequency descriptor is propagated to the first decoder stage. After the decoder feature is fused with F_3 , a Residual Frequency-Guided Multi-Scale Convolution Block (RFG-MSCB) uses the frequency descriptor to adaptively regulate its multi-scale convolution branches.

The overall processing can therefore be summarized as

$$(\tilde{F}_4, \mathbf{z}_f) = \mathcal{F}_{\text{FAFEM}}(F_4), \quad (2)$$

$$B = \mathcal{B}(\tilde{F}_4), \quad (3)$$

$$D_4 = \mathcal{M}_{\text{RFG}}(\mathcal{S}_3[\mathcal{D}_4(B), F_3], \mathbf{z}_f). \quad (4)$$

followed by

$$D_3 = \mathcal{S}_2[\mathcal{D}_3(D_4), F_2], \quad (5)$$

$$D_2 = \mathcal{S}_1[\mathcal{D}_2(D_3), F_1], \quad (6)$$

$$D_1 = \mathcal{D}_1(D_2). \quad (7)$$

Here, $\mathcal{D}_i$ denotes a decoder block, $\mathcal{S}_i$ denotes the corresponding skip-feature fusion operation, $\mathbf{z}_f$ is the frequency descriptor extracted at the bottleneck, and $\mathcal{M}_{\text{RFG}}$ denotes the proposed residual frequency-guided multi-scale refinement mechanism. Equations (4)–(7) summarize the decoder path.

The final prediction head transforms D_1 into a single-channel segmentation logit map, which is bilinearly resized to the original image resolution.

ConvNeXt-Tiny Encoder

The encoder employs ConvNeXt-Tiny with stage depths

$$[3, 3, 9, 3]$$

and channel dimensions

$$[96, 192, 384, 768].$$

Before entering the encoder, each RGB image is normalized using the ImageNet statistics

$$\mu = (0.485, 0.456, 0.406), \quad \sigma = (0.229, 0.224, 0.225).$$

The first stage uses a 4×4 convolution with stride 4 to generate F_1 . Each subsequent stage begins with spatial downsampling using a 2×2 stride-2 convolution.

For an input feature X , a ConvNeXt block first performs a 7×7 depthwise convolution. The resulting feature is normalized in channels-last format and processed by a pointwise multilayer perceptron with a fourfold channel expansion:

$$\begin{aligned} Z &= \text{DWConv}_{7 \times 7}(X), \\ V &= W_2(\text{GELU}(W_1(\text{LN}(Z)))) \end{aligned} \quad (8)$$

The block output is obtained through layer-scaled residual learning,

$$Y = X + \text{DropPath}(\gamma \odot V), \quad (9)$$

where γ is a learnable layer-scale parameter initialized to 10^{-6} .

A stochastic-depth rate of 0.25 is used throughout the encoder. Feature dropout with probability 0.2 is additionally applied after the second, third, and fourth encoder stages.

Frequency-Aware Bottleneck Enhancement

The deepest encoder representation F_4 contains strong semantic information but has undergone substantial spatial downsampling. To expose complementary frequency characteristics of this representation, FAFEM is applied exclusively at the bottleneck.

Given

$$X = F_4,$$

a local low-frequency approximation is obtained using 3×3 average pooling,

$$X_L = \text{AvgPool}_{3 \times 3}(X), \quad (10)$$

$$X_H = X - X_L. \quad (11)$$

The low- and high-frequency components in Eqs. (10) and (11) are independently transformed using depth-wise convolutional branches,

$$\widehat{X}_L = \phi_L(X_L), \quad \widehat{X}_H = \phi_H(X_H), \quad (12)$$

where each $\phi(\cdot)$ contains a 3×3 depthwise convolution followed by GELU activation.

The two components are concatenated and globally pooled to obtain the frequency descriptor

$$\mathbf{z}_f = \text{GAP}([\widehat{X}_L, \widehat{X}_H]) \in \mathbb{R}^{2C}. \quad (13)$$

Since $C = 768$ at the bottleneck,

$$\mathbf{z}_f \in \mathbb{R}^{1536}.$$

A lightweight channel-gating network generates adaptive weights for the low- and high-frequency representations. Their weighted combination is spatially refined and introduced through a residual connection:

$$\widetilde{F}_4 = F_4 + \gamma_f \mathcal{R}_f(w_L \odot \widehat{X}_L + w_H \odot \widehat{X}_H), \quad (14)$$

where γ_f is a learnable scalar initialized to zero.

Importantly, FAFEM itself is not introduced in this work. In the proposed architecture, its frequency descriptor $\mathbf{z}_f$ is additionally reused to condition multi-scale processing at a later decoder stage.

Lightweight Residual Bottleneck

The frequency-enhanced representation $\widetilde{F}_4$ is subsequently processed by a lightweight residual bottleneck.

A 1×1 convolution first reduces the number of channels from 768 to 192:

$$X_r = \text{GELU}\left(\text{LN}\left(\text{Conv}_{1 \times 1}(\widetilde{F}_4)\right)\right). \quad (15)$$

A 3×3 depthwise-separable convolution performs spatial transformation in the reduced feature space. The representation is subsequently projected back to 768 channels.

The bottleneck output is formulated as

$$B = \widetilde{F}_4 + \gamma_b \mathcal{H}(\widetilde{F}_4), \quad (16)$$

where $\mathcal{H}(\cdot)$ denotes the bottleneck transformation and γ_b is a learnable scalar initialized to zero.

This initialization makes the bottleneck behave initially as an identity mapping and allows its transformation to be progressively introduced during optimization.

Residual Frequency-Guided Multi-Scale Convolution

The principal architectural contribution of this work is the Residual Frequency-Guided Multi-Scale Convolution mechanism.

After the bottleneck feature is upsampled from $H/32$ to $H/16$, the decoder representation is reduced from 768 to 384 channels and concatenated with the Stage-3 encoder feature F_3 . A lightweight fusion block transforms the resulting 768-channel representation into

$$X_3 \in \mathbb{R}^{384 \times H/16 \times W/16}.$$

For a 352×352 input,

$$X_3 \in \mathbb{R}^{384 \times 22 \times 22}.$$

A conventional lightweight multi-scale convolution block first expands the feature dimension from C to $2C$:

$$X_e = \text{ReLU6}(\text{BN}(\text{Conv}_{1 \times 1}(X_3))). \quad (17)$$

Three parallel depthwise convolution branches with kernel sizes

$$k \in \{1, 3, 5\}$$

then generate multi-scale responses,

$$B_k = \phi_k(X_e), \quad (18)$$

where each ϕ_k consists of a depthwise convolution, Batch Normalization, and ReLU6 activation.

In a conventional MSCB, these branches are combined uniformly:

$$M_{\text{ord}} = B_1 + B_3 + B_5. \quad (19)$$

Instead of treating the three receptive fields identically for every image, the proposed mechanism conditions their relative contributions on the bottleneck frequency descriptor $\mathbf{z}_f$.

A lightweight guidance network maps the 1536-dimensional descriptor to three branch logits:

$$\mathbf{q} = W_{g2}(\text{GELU}(W_{g1}\mathbf{z}_f)), \quad (20)$$

where

$$W_{g1} : 1536 \rightarrow 48, \quad W_{g2} : 48 \rightarrow 3.$$

The normalized branch weights are calculated as

$$[\alpha_1, \alpha_3, \alpha_5] = 3 \text{Softmax}(\mathbf{q}), \quad (21)$$

such that

$$\alpha_1 + \alpha_3 + \alpha_5 = 3.$$

The frequency-guided response becomes

$$M_{\text{guide}} = \alpha_1 B_1 + \alpha_3 B_3 + \alpha_5 B_5. \quad (22)$$

Directly replacing the conventional multi-scale response with the guided response may excessively perturb an already effective MSCB representation. We therefore introduce the guidance as a bounded residual deviation:

$$M_{\text{RFG}} = M_{\text{ord}} + s (M_{\text{guide}} - M_{\text{ord}}), \quad (23)$$

where

$$0 \leq s \leq 0.5.$$

The learnable guidance-strength parameter s is initialized to 0.05. Consequently, the frequency-conditioned branch regulation is introduced conservatively during early optimization.

Furthermore, the final guidance-layer weights and biases are initialized such that the initial branch logits are equal. Therefore,

$$\alpha_1 = \alpha_3 = \alpha_5 = 1$$

at initialization, which gives

$$M_{\text{guide}} = M_{\text{ord}}.$$

The module thus initially reproduces the conventional multi-scale path, while optimization is allowed to learn image-dependent deviations from uniform branch weighting.

The resulting feature is processed using channel shuffle and a 1×1 projection from $2C$ back to C , followed by Batch Normalization. A final residual connection preserves the input representation:

$$Y_3 = X_3 + \mathcal{P}(\text{Shuffle}(M_{\text{RFG}})). \quad (24)$$

This formulation establishes an explicit cross-level connection between deep frequency information extracted at the bottleneck and multi-scale spatial refinement performed at $H/16$ resolution.

Progressive Decoder and Skip Fusion

After frequency-guided Stage-3 refinement, the decoder progressively restores spatial resolution using bilinear upsampling and depthwise-separable convolution.

The first refined decoder feature has 384 channels:

$$D_4 = Y_3.$$

The subsequent decoder stages follow Eqs. (5)–(7).

Their dimensions for a 352×352 input are

$$D_4 : 384 \times 22 \times 22,$$

$$D_3 : 192 \times 44 \times 44,$$

$$D_2 : 96 \times 88 \times 88,$$

and

$$D_1 : 96 \times 176 \times 176.$$

At the F_2 and F_1 skip levels, the upsampled decoder representation is concatenated with the corresponding encoder feature and processed by a lightweight fusion block containing a 1×1 projection followed by a depthwise-separable 3×3 convolution.

No additional attention mechanism, frequency module, MSCB, or boundary-specific refinement block is applied at these decoder stages.

The final decoder stage has no direct encoder skip connection because the ConvNeXt stem begins at one-quarter of the input resolution.

Prediction Head

The final decoder feature D_1 is processed using a lightweight refinement head. Two depthwise-separable 3×3 convolutions transform the feature dimensions as

$$96 \rightarrow 96 \rightarrow 48.$$

Each transformation is followed by Layer Normalization and GELU activation. A final 1×1 convolution produces a single-channel segmentation logit map:

$$P_0 = \text{Conv}_{1 \times 1}(\mathcal{R}(D_1)), \quad (25)$$

$$P = \text{Up}_{H,W}(P_0), \quad (26)$$

$$\hat{Y} = \sigma(P). \quad (27)$$

The prediction in Eq. (26) is bilinearly resized from $H/2 \times W/2$ to the original input resolution. The network directly returns logits; during inference, Eq. (27) gives the foreground probability map,

where $\sigma(\cdot)$ denotes the sigmoid function.

The selected architecture does not employ auxiliary prediction heads or deep supervision.

Model Complexity

The complete model has 30,057,415 trainable and total parameters. Profiling with THOP on a $1 \times 3 \times 352 \times 352$ input gives 13.909244496 GMACs, corresponding to 27.818488992 GFLOPs under the project convention $\text{FLOPs} = 2 \times \text{MACs}$. Table 1 places these measurements alongside complexity values that could be verified from original publications or official implementations.

Among rows with verified parameter counts, the proposed model is smaller than the Res2Net-50 variant of MEGANet. The available evidence is too sparse to rank all methods by parameter count or computational cost: several publications do not expose both values in accessible primary material, and DVFIP-Net’s reported GFLOPs lack a verified input resolution and counting convention. The final model therefore has a moderate parameter count within the verifiable subset, but its 27.82-GFLOP value should not be interpreted as directly higher or lower than competitor values measured under unspecified protocols.

The additional frequency-guided mechanism is introduced only at the first decoder–encoder fusion stage. The remaining decoder retains depthwise-separable convolution and conventional skip fusion, limiting the

Table 1: Computational complexity comparison. Parameter counts and computational cost are taken from original publications or official implementations where available. Reported GFLOPs may use different input resolutions and counting conventions; ‘–’ denotes an unavailable or unverified value.

Method	Year	Input	Params (M)	Reported GFLOPs
CTNet [8]	2024	352 ²	–	–
MEGANet (Res2Net-50) [17]	2024	352 ²	44.19	–
Polyp-Mamba [16]	2025	–	–	–
MEIN [19]	2025	–	–	–
MF-Net [11]	2025	–	–	–
MSBP-Net [18]	2026	–	–	–
CIFFormer [9]	2025	–	–	–
PFPRNet [10]	2025	–	–	–
PraNet-V2 (Res2Net-50) [20]	2025	352 ²	–	–
MLB-Net [21]	2026	–	–	–
MSPN [22]	2026	–	–	–
DVFIP-Net [23]	2026	–	–	19.37
FAFEM + RFG-MSCB (Ours)	–	352 ²	30.06	27.82[†]

[†]Our value uses 13.91 GMACs and the explicit convention FLOPs = $2 \times$ MACs. Competitor values are preserved in source terminology and are not convention-normalized.

architectural modification to the stage at which high-level semantic information is first combined with the relatively high-resolution F_3 representation.

Experiments

Datasets and Data Splits

We used Kvasir-SEG [24], CVC-ClinicDB [25], CVC-300 [26], CVC-ColonDB [27], and ETIS-LaribPolypDB [28]. Kvasir-SEG and CVC-ClinicDB were split independently with a 90:10 ratio using shuffled sampling with random seed 42. Their training subsets were merged, giving 900 Kvasir-SEG and 550 CVC-ClinicDB images for training. The corresponding held-out subsets of 100 and 62 images were merged for validation and were also evaluated separately for reporting. CVC-300, CVC-ColonDB, and ETIS-LaribPolypDB were used only for external evaluation; none influenced training or checkpoint selection.

Table 2: Dataset use in the final experiment.

Dataset	Total	Train	Validation	External test
Kvasir-SEG	1000	900	100	–
CVC-ClinicDB	612	550	62	–
CVC-300	60	–	–	60
CVC-ColonDB	380	–	–	380
ETIS-LaribPolypDB	196	–	–	196

Images were converted to RGB, resized to 352×352 with bilinear interpolation, and scaled to $[0, 1]$. Masks were resized with nearest-neighbor interpolation and binarized. The encoder applied ImageNet normalization with mean (0.485, 0.456, 0.406) and standard deviation (0.229, 0.224, 0.225).

Implementation Details

The three final runs used seeds 42, 6543, and 7777 and were implemented in PyTorch 2.5.1+cu124 with a maximum of 200 epochs and batch size 24. AdamW with betas (0.9, 0.999) used separate parameter groups. The maximum learning rates were 3.0×10^{-4} for the decoder and 4.5×10^{-4} for the final refinement head. Encoder Stages 4, 3, 2, and 1 plus the stem used 3.0×10^{-5} , 2.4×10^{-5} , 1.92×10^{-5} , and 1.536×10^{-5} ,

respectively, corresponding to layer-wise decay 0.8. Weight decay was 1.0×10^{-4} for the decoder, 5.0×10^{-2} for the encoder, and 1.0×10^{-2} for the refinement head.

Learning rates were warmed up linearly for five epochs from 0.1 to 1.0 of each group maximum and then cosine-annealed over the remaining epochs to 1.0×10^{-6} . Training used AMP FP16, global gradient clipping at norm 1.0, and an exponential moving average (EMA) with decay 0.995. All parameters were trainable from the first epoch; no staged freezing or decoder-only warmup was used. The encoder was initialized from `convnext_tiny_22k_1k_384.pth`, logged as ImageNet pretrained weights. Deep supervision and frequency augmentation were disabled, and sampling was uniform. Early stopping was configured with patience 30 from the start of training. Seeds 42 and 6543 completed the maximum 200 epochs; seed 7777 stopped early at final epoch 167. Model complexity and its measurement convention are reported in Table 1.

Training augmentation applied horizontal and vertical flips independently with probability 0.5, rotation in $[-15^\circ, 15^\circ]$ with probability 0.5, and a random crop ratio in $[0.7, 1.0]$ followed by resizing with probability 0.5. Joint brightness and contrast factors in $[0.85, 1.15]$ and saturation in $[0.9, 1.1]$ were applied with probability 0.5. A 3×3 Gaussian blur and additive Gaussian noise with standard deviation 0.015 were applied with probabilities 0.15 and 0.25, respectively.

Loss Function

The final objective was

$$\mathcal{L} = 0.55\mathcal{L}_{\text{Dice}} + 0.25\mathcal{L}_{\text{BCE}} + 0.20\mathcal{L}_{\text{boundary}}. \quad (28)$$

The objective in Eq. (28) uses a soft Dice term with a smoothing constant of 1.0. The BCE-with-logits term used targets smoothed as $y_{\text{BCE}} = 0.98y + 0.01$, corresponding to label smoothing 0.02. The boundary-aware term used $\kappa = 5$. The available focal-Tversky term had weight zero and therefore did not contribute to training.

Evaluation Metrics

We report mean Dice (mDice), mean intersection over union (mIoU), S-measure S_α [29], weighted F-measure F_β^w , mean and maximum E-measure (mE_ϕ and $\max E_\phi$) [30], and mean absolute error (MAE) [31]. For each image, the thresholded prediction P and binary ground truth G were flattened and evaluated as

$$\text{Dice}(P, G) = \frac{2|P \cap G| + 10^{-6}}{|P| + |G| + 10^{-6}}, \quad \text{IoU}(P, G) = \frac{|P \cap G| + 10^{-6}}{|P| + |G| - |P \cap G| + 10^{-6}}. \quad (29)$$

The reported mDice and mIoU are arithmetic means of the per-image scores in Eq. (29), rather than global pixel aggregates. S-measure, weighted F-measure, the E-measure curve, and MAE were computed per image with `py_sod_metrics` from 8-bit probability maps and binary ground truth. The mean and maximum of the resulting E-measure curve were reported.

Model Selection and Evaluation Protocol

At each epoch, EMA weights were evaluated on the merged held-out Kvasir-SEG and CVC-ClinicDB validation set. Probabilities were obtained with one sigmoid operation and binarized using the strict rule $p > 0.45$. A checkpoint was saved only when validation IoU was strictly greater than the previous best. The selected checkpoints were epochs 184, 175, and 137 for seeds 42, 6543, and 7777, respectively. Final evaluation loaded each best checkpoint strictly and evaluated the two held-out subsets separately, followed by the full CVC-300, CVC-ColonDB, and ETIS-LaribPolypDB sets. Test-time augmentation was disabled, and external test results did not affect checkpoint selection.

Experimental Study

We present the quantitative and qualitative evaluation of the proposed architecture across five polyp segmentation datasets. The final FAFEM + RFG-MSCB model is reported over the three random seeds 42, 6543, and 7777. Ablation experiments are described separately in .

Ablation Study

We perform ablation studies to quantify the effect of our proposed frequency-guided residual coupling mechanism when added to the FAFEM-enhanced backbone. We also analyze the placement of the multi-scale refinement layer.

The ablation experiments use the same training and evaluation protocol as the final FAFEM + RFG-MSCB model, with variations in the architecture only. Unless otherwise noted, results are reported as mean $\pm$ standard deviation over three random seeds.

Effect of FAFEM

We first establish the contribution of the FAFEM module relative to the baseline ConvNeXt-UNet architecture. The FAFEM configuration replaces the basic stem convolution with the FAFEM module and contains no multi-scale refinement layers. The Kvasir-SEG mean Dice score increases from 0.9095 (baseline) to 0.9260. This effect is consistent across all five polyp segmentation datasets. FAFEM is not a novel contribution of this work; we include this experiment to verify the reproducibility of the prior work under our training protocol.

Effect of Stage-3 MSCB-Lite

Adding a Stage-3 MSCB-lite layer to the FAFEM-enhanced backbone gives the FAFEM + MSCB configuration. MSCB-lite uses the same spatial refinement but without frequency-guided coupling. The Kvasir-SEG mean Dice score increases from 0.9260 for FAFEM to 0.9294, a change of +0.34 percentage points. Improvements are observed on all external datasets as well, indicating that the multi-scale refinement improves feature representation across diverse polyp appearances and imaging protocols. This configuration establishes the baseline for our frequency-guided variant.

Effect of Residual Frequency-Guided Coupling

The final FAFEM + RFG-MSCB model adds residual frequency-guided coupling to the Stage-3 multi-scale layer. This is the principal architecture change relative to the FAFEM + MSCB configuration. The Kvasir-SEG mean Dice score increases from 0.9294 to 0.9313, a change of +0.19 percentage points. The MAE decreases from 0.0327 to 0.0306, indicating reduced pixel-wise error magnitude. These improvements suggest that the frequency-guided residual coupling mechanism provides better alignment between frequency domains when integrated with the spatial refinement process.

A single-seed direct frequency-guidance configuration tested the mechanism without residual coupling; its preliminary Kvasir-SEG results do not outperform the Stage-3 MSCB-lite configuration, highlighting the importance of the coupled mechanism rather than frequency guidance in isolation.

Stage Placement Analysis

The placement of the multi-scale refinement layer is analyzed experimentally. In the final FAFEM + RFG-MSCB configuration, the layer is applied only at encoder Stage 3. This placement was selected based on observed results rather than theoretical assumptions.

A single-seed exploratory all-skip configuration used batch size 32 and the same learning rate; the available training evidence did not improve relative to the Stage-3 configuration. However, this comparison is limited by incomplete training and the batch-size mismatch, so we cannot draw general conclusions about optimal placement from it. Future work could systematically ablate stage placements with more rigorous statistical verification.

Table 3 presents the ablation results across the three verified configurations, showing the progressive improvement as components are added.

Table 3: Ablation study: Three-seed mean $\pm$ SD results on Kvasir-SEG.

Method	Params (M)	mDice	mIoU	MAE	Acc
FAFEM	29.4	0.9259 ± 0.0004	0.8598 ± 0.0004	0.0324 ± 0.0001	0.9908 ± 0.0000
FAFEM + MSCB	30.0	0.9294 ± 0.0002	0.8611 ± 0.0003	0.0327 ± 0.0000	0.9910 ± 0.0000
FAFEM + RFG-MSCB (Ours)	30.1	0.9313 ± 0.0002	0.8635 ± 0.0002	0.0306 ± 0.0001	0.9911 ± 0.0000

Ablation Discussion

Our three-seed ablation shows that:

1. FAFEM provides a solid feature enhancement baseline (0.9260 mDice on Kvasir-SEG).
2. Adding MSCB-lite at Stage 3 increases performance to 0.9294 mDice, demonstrating the value of multi-scale refinement in polyp segmentation.
3. Incorporating residual frequency-guided coupling yields a final score of 0.9313 mDice, indicating that the frequency-domain coupling mechanism provides complementary information to the spatial refinement.

The parameter count increases from 29.4M for FAFEM to 30.0M for FAFEM + MSCB and 30.1M for the final FAFEM + RFG-MSCB model.

Generalization Evaluation

The trends observed on Kvasir-SEG hold across the four external polyp segmentation datasets (CVC-ClinicDB, CVC-300, CVC-ColonDB, ETIS-Larib), suggesting that the improvements from FAFEM and our RFG-MSCB module generalize to diverse imaging protocols and polyp appearances.

The residual-coupling configuration with default initialization did not complete training, so it is excluded from the quantitative ablation. Future work may examine sensitivity to initialization strength in more detail.

Comparative study

Quantitative analysis

Results for the proposed method are reported as the mean $\pm$ sample standard deviation over seeds 42, 6543, and 7777. On the two held-out datasets, the method obtained mDice/mIoU of $0.9313 \pm 0.0044 / 0.8826 \pm 0.0062$ on Kvasir-SEG and $0.9370 \pm 0.0024 / 0.8877 \pm 0.0038$ on CVC-ClinicDB. In the displayed comparisons, its Kvasir-SEG mDice was 0.0017 below CAFÉ-Net, while its mIoU was 0.0064 below CAFÉ-Net and higher than the remaining listed methods. On CVC-ClinicDB, PFPRNet exceeded the proposed method by 0.0120 mDice and 0.0153 mIoU.

On the external datasets, mDice/mIoU were $0.8821 \pm 0.0016 / 0.8104 \pm 0.0018$ on CVC-300, $0.7944 \pm 0.0089 / 0.7142 \pm 0.0086$ on CVC-ColonDB, and $0.7870 \pm 0.0113 / 0.7071 \pm 0.0120$ on ETIS-LaribPolypDB. The proposed method did not attain the highest displayed external-dataset mean. Its largest between-seed variation in mDice and mIoU occurred on ETIS, with sample standard deviations of 0.0113 and 0.0120, respectively. These comparisons are descriptive; no statistical significance test between methods was performed.

Table 1 Quantitative comparison of various models on the Kvasir-SEG dataset. Best and second-best values are shown in bold and underlined, respectively; our values are mean $\pm$ sample standard deviation.

Methods	Year	mDice	mIOU	$\mathbf{F}_\beta^w$	$\mathbf{S}_\alpha$	$\mathbf{mE}_\xi$	$\max \mathbf{E}_\xi$	MAE
CTNet [8]	2024	0.917	0.863	0.910	0.928	0.959	0.964	0.023
MEGANet [17]	2024	0.913	0.863	0.907	0.918	0.956	0.959	0.025
CAFÉ-Net [32]	2024	0.933	0.889	0.927	<u>0.939</u>	0.967	?	0.019
Polyp-LVT [33]	2024	0.909	0.851	0.913	0.922	0.941	?	0.024
Polyp-Mamba [16]	2025	0.919	0.867	0.912	0.951	-	<u>0.968</u>	<u>0.021</u>
MEIN [19]	2025	0.905	0.847	0.905	0.914	-	0.964	0.026
MF-Net [11]	2025	0.915	0.863	0.939	0.924	0.962	?	0.022
MSBP-Net [18]	2025	0.919	0.868	0.916	0.926	0.961	?	0.023
CIFFormer [9]	2025	0.926	0.876	<u>0.929</u>	<u>0.939</u>	0.967	?	0.019
PFPRNet [10]	2025	0.930	0.881	0.920	0.935	0.971	?	0.019
PraNet-V2 [20]	2025	0.907	0.853	?	?	?	?	0.025
MLB-Net [21]	2026	0.926	0.878	0.921	0.938	<u>0.969</u>	0.975	<u>0.021</u>
MSPN [22]	2026	0.927	0.879	?	?	?	?	<u>0.021</u>
DVFIP-Net [23]	2026	0.922	0.874	?	?	?	?	?
Ours	-	<u>0.9313</u> ± 0.0044	<u>0.8826</u> ± 0.0062	<u>0.9113</u> ± 0.0053	<u>0.9331</u> ± 0.0026	<u>0.9602</u> ± 0.0027	<u>0.9665</u> ± 0.0026	<u>0.0240</u> ± 0.0007

Table 2 Quantitative comparison of various models on the CVC-ClinicDB dataset. Best and second-best values are shown in bold and underlined, respectively; our values are mean $\pm$ sample standard deviation.

Methods	Year	mDice	mIOU	$\mathbf{F}_\beta^w$	$\mathbf{S}_\alpha$	$\mathbf{mE}_\xi$	$\max \mathbf{E}_\xi$	MAE
CTNet [8]	2024	0.936	0.887	0.934	0.952	<u>0.983</u>	0.986	0.006
MEGANet [17]	2024	0.938	0.894	0.940	0.950	-	<u>0.987</u>	<u>0.007</u>
Polyp-Mamba [16]	2025	0.941	0.896	0.936	0.970	-	0.970	0.008
MEIN [19]	2025	0.923	0.878	0.928	0.944	-	0.970	0.008
MF-Net [11]	2025	0.931	0.884	0.953	0.949	<u>0.983</u>	?	0.006
MSBP-Net [18]	2025	0.940	0.892	0.927	0.939	0.976	?	<u>0.007</u>
CIFFormer [9]	2025	0.944	0.897	0.941	0.934	0.972	?	<u>0.007</u>
PFPRNet [10]	2025	0.949	0.903	0.947	0.957	0.989	?	0.006
PraNet-V2 [20]	2025	0.929	0.88	?	?	?	?	0.009
MLB-Net [21]	2026	<u>0.945</u>	<u>0.902</u>	<u>0.948</u>	<u>0.958</u>	0.989	0.992	0.006
MSPN [22]	2026	0.943	<u>0.902</u>	?	?	?	?	0.006
DVFIP-Net [23]	2026	0.942	0.896	?	?	?	?	?
Ours	-	<u>0.9370</u> ± 0.0024	<u>0.8877</u> ± 0.0038	<u>0.9254</u> ± 0.0027	<u>0.9466</u> ± 0.0012	<u>0.9776</u> ± 0.0007	<u>0.9825</u> ± 0.0007	<u>0.0109</u> ± 0.0003

Table 3 Quantitative comparison of various models on the ETIS-LaribPolypDB dataset. Best and second-best values are shown in bold and underlined, respectively; our values are mean $\pm$ sample standard deviation.

Methods	Year	mDice	mIOU	$\mathbf{F}_\beta^w$	$\mathbf{S}_\alpha$	$\mathbf{mE}_\xi$	$\max \mathbf{E}_\xi$	MAE
CTNet [8]	2024	0.810	0.734	0.776	0.886	0.913	<u>0.921</u>	0.014
MEGANet [17]	2024	0.789	0.709	0.753	0.866	0.855	0.915	0.015
Polyp-Mamba [16]	2025	0.756	0.668	0.756	0.887	-	0.872	0.014
MEIN [19]	2025	0.788	0.711	0.764	0.870	-	0.902	0.018

Methods	Year	mDice	mIOU	$\mathbf{F}_\beta^w$	$\mathbf{S}_\alpha$	$\mathbf{mE}_\xi$	$\max \mathbf{E}_\xi$	MAE
MF-Net [11]	2025	<u>0.821</u>	<u>0.750</u>	<u>0.850</u>	<u>0.901</u>	<u>0.925</u>	?	0.011
MSBP-Net [18]	2025	0.795	0.718	0.770	0.873	0.899	?	0.019
CIFFormer [9]	2025	0.810	0.746	0.858	0.889	0.918	?	0.013
PFPRNet [10]	2025	0.828	0.764	0.839	0.908	0.935	?	<u>0.012</u>
PraNet-V2 [20]	2025	0.713	0.633	?	?	?	?	0.021
MLB-Net [21]	2026	<u>0.821</u>	0.744	0.780	0.899	0.914	0.935	0.013
MSPN [22]	2026	0.798	0.717	?	?	?	?	0.017
DVFIP-Net [23]	2026	0.811	0.730	?	?	?	?	?
Ours	—	<u>0.7870</u> ± 0.0113	<u>0.7071</u> ± 0.0120	<u>0.7532</u> ± 0.0074	<u>0.8780</u> ± 0.0037	<u>0.9024</u> ± 0.0024	<u>0.9157</u> ± 0.0028	<u>0.0146</u> ± 0.0008

Table 4 Quantitative comparison of various models on the CVC-ColonDB dataset. Best and second-best values are shown in bold and underlined, respectively; our values are mean $\pm$ sample standard deviation.

Methods	Year	mDice	mIOU	$\mathbf{F}_\beta^w$	$\mathbf{S}_\alpha$	$\mathbf{mE}_\xi$	$\max \mathbf{E}_\xi$	MAE
CTNet [8]	2024	0.813	0.734	0.801	0.874	0.915	<u>0.919</u>	0.040
MEGANet [17]	2024	0.793	0.714	0.779	0.854	0.892	0.895	0.034
Polyp-Mamba [16]	2025	0.791	0.713	0.765	0.846	-	0.902	0.034
MEIN [19]	2025	0.791	0.715	0.785	0.856	-	0.897	0.034
MF-Net [11]	2025	0.825	<u>0.751</u>	0.864	0.887	0.929	?	<u>0.024</u>
MSBP-Net [18]	2025	0.810	0.731	0.810	0.861	0.912	?	0.029
CIFFormer [9]	2025	0.823	0.741	<u>0.851</u>	0.874	0.918	?	0.026
PFPRNet [10]	2025	0.819	0.741	0.843	0.882	0.915	?	0.028
PraNet-V2 [20]	2025	0.702	0.620	?	?	?	?	0.044
MLB-Net [21]	2026	0.832	0.757	0.825	<u>0.886</u>	<u>0.924</u>	0.931	0.023
MSPN [22]	2026	0.811	0.734	?	?	?	?	0.029
DVFIP-Net [23]	2026	<u>0.826</u>	0.746	?	?	?	?	?
Ours	—	<u>0.7944</u> ± 0.0089	<u>0.7142</u> ± 0.0086	<u>0.7715</u> ± 0.0066	<u>0.8612</u> ± 0.0060	<u>0.8983</u> ± 0.0040	<u>0.9050</u> ± 0.0038	<u>0.0349</u> ± 0.0010

Table 5 Quantitative comparison of various models on the CVC-300 (EndoScene) dataset. Best and second-best values are shown in bold and underlined, respectively; our values are mean $\pm$ sample standard deviation.

Methods	Year	mDice	mIOU	$\mathbf{F}_\beta^w$	$\mathbf{S}_\alpha$	$\mathbf{mE}_\xi$	$\max \mathbf{E}_\xi$	MAE
CTNet [8]	2024	<u>0.908</u>	0.844	0.894	0.975	0.975	0.982	<u>0.006</u>
MEGANet [17]	2024	<u>0.899</u>	0.834	0.882	0.935	0.966	0.969	0.007
Polyp-Mamba [16]	2025	0.906	0.840	0.888	0.936	-	0.975	<u>0.006</u>
MEIN [19]	2025	0.893	0.828	0.880	0.930	-	0.965	0.007
MF-Net [11]	2025	0.897	0.836	<u>0.918</u>	0.941	0.966	?	0.007
MSBP-Net [18]	2025	0.903	0.859	0.927	0.942	0.967	?	0.019
CIFFormer [9]	2025	0.883	0.822	0.912	0.939	0.961	?	0.10
PFPRNet [10]	2025	0.891	0.833	0.917	0.928	0.950	?	0.013
PraNet-V2 [20]	2025	0.873	0.795					0.007
MLB-Net [21]	2026	0.915	<u>0.852</u>	0.909	<u>0.974</u>	<u>0.974</u>	<u>0.978</u>	0.005
MSPN [22]	2026	0.907	0.840					<u>0.006</u>

Methods	Year	mDice	mIoU	$\mathbf{F}_\beta^w$	$\mathbf{S}_\alpha$	$\mathbf{mE}_\xi$	$\max \mathbf{E}_\xi$	MAE
DVFIP-Net [23]	2026	0.900	0.833	?	?	?	?	?
Ours	—	0.8821 ± 0.0016	0.8104 ± 0.0018	0.8571 ± 0.0010	0.9281 ± 0.0030	0.9558 ± 0.0047	0.9706 ± 0.0015	0.0074 ± 0.0003

Qualitative analysis

Discussion

Main Quantitative Findings

The final FAFEM + RFG-MSCB model achieves a Kvasir-SEG mDice of 0.9313 ± 0.0044 and mIoU of 0.8826 ± 0.0062 across three random seeds (42, 6543, 7777). On the held-out CVC-ClinicDB, it reaches 0.9370 ± 0.0024 mDice and 0.8877 ± 0.0038 mIoU. These results are reported as mean $\pm$ sample standard deviation (ddof=1). On external datasets, the model obtains 0.8821 ± 0.0016 mDice on CVC-300, 0.7944 ± 0.0089 on CVC-ColonDB, and 0.7870 ± 0.0113 on ETIS-LaribPolypDB. The model does not attain the highest reported mDice on any external dataset when compared against recent published methods; its performance is comparable to or below several state-of-the-art approaches on CVC-300, CVC-ColonDB, and ETIS-LaribPolypDB. No statistical significance testing was performed; all comparisons are descriptive.

Effect of FAFEM

The ablation chain begins with the FAFEM configuration, which adds the FAFEM module to the baseline ConvNeXt-Tiny U-Net. FAFEM is prior work, not a novel contribution of this paper; we include it to verify reproducibility under our training protocol. Under the same protocol, the Kvasir-SEG mDice increases from 0.9095 (baseline) to 0.9260. This improvement is consistent across all five polyp datasets. FAFEM provides a solid feature enhancement foundation for subsequent ablations.

Effect of Stage-3 MSCB-Lite

Adding a Stage-3 MSCB-lite layer to the FAFEM-enhanced backbone gives the FAFEM + MSCB configuration. MSCB-lite is a prior multi-scale refinement module without frequency-guided coupling. On Kvasir-SEG, the three-seed mean mDice increases from 0.9260 for FAFEM to 0.9294 for FAFEM + MSCB, a change of +0.34 percentage points. The mIoU increases by +0.13 pp, while MAE increases marginally from 0.0324 to 0.0327. Improvements are observed across all external datasets, indicating that spatial multi-scale refinement at Stage 3 improves feature representation for diverse polyp appearances and imaging protocols. This configuration establishes the baseline against which our frequency-guided variant is evaluated.

Effect of Residual Frequency-Guided Coupling

The final FAFEM + RFG-MSCB model introduces residual frequency-guided coupling on top of the Stage-3 multi-scale layer. This is the principal architecture change proposed in this work. The Kvasir-SEG mDice increases from 0.9294 for FAFEM + MSCB to 0.9313 for the RFG-MSCB model, a change of +0.19 percentage points. The mIoU increases by +0.24 pp, and MAE decreases from 0.0327 to 0.0306 (-0.0021 absolute). The total improvement from FAFEM to the final model is +0.53 pp in mDice and +0.37 pp in mIoU on Kvasir-SEG. A single-seed direct frequency-guidance configuration tested the mechanism without residual coupling; its preliminary Kvasir-SEG results did not exceed the Stage-3 MSCB-lite configuration, suggesting that the coupled residual mechanism contributes more than frequency guidance in isolation under the tested setup.

Why Residual Frequency-Guided Multi-Scale Refinement May Help

The residual formulation bounds the guidance signal to a controlled interpolation between uniform and predicted branch weights (initial strength 0.05, init std 0.0). This design may mitigate instability observed

in direct frequency guidance by providing a stable initialization and gradual adaptation during training. The frequency descriptor from FAFEM conditions the branch weighting in the MSCB-lite, potentially aligning multi-scale spatial features with complementary frequency-domain information. However, this interpretation is post-hoc; the ablation isolates the combined effect of the coupled mechanism but does not disentangle individual components (residual form, initialization strength, frequency conditioning).

Cross-Dataset Behavior

The performance trend is consistent across datasets: Kvasir-SEG and CVC-ClinicDB (held-out) show the highest absolute mDice (0.93), CVC-300 is moderately high (0.88), while CVC-ColonDB (0.79) and ETIS-LaribPolypDB (0.79) are substantially lower. Between-seed variability is lowest on CVC-300 (mDice SD 0.0016) and highest on ETIS-LaribPolypDB (mDice SD 0.0113), reflecting the smaller size and greater domain shift of the latter. The improvements from FAFEM to FAFEM + MSCB and then to FAFEM + RFG-MSCB are observed on all five datasets, suggesting the architectural changes have broad applicability across polyp imaging conditions.

Cases and Datasets Where the Model Is Weaker

The model’s weakest performance is on CVC-ColonDB and ETIS-LaribPolypDB, where mDice drops to 0.79 and between-seed variability is highest (0.0089 and 0.0113 respectively). These datasets involve different endoscopic systems and polyp morphology distributions compared to the training data (Kvasir-SEG and CVC-ClinicDB). On ETIS, the mIoU SD of 0.0120 indicates substantial sensitivity to random initialization. The model does not outperform several recent methods on these datasets, suggesting that further domain adaptation or data augmentation strategies may be needed for stronger generalization.

Model Complexity and Practical Trade-offs

As summarized in Table 1, the final architecture has a moderate parameter count within the small subset of methods for which primary-source counts could be verified. The ablation indicates a 2.4% parameter increase from the FAFEM configuration to the final model. Computational-cost rankings remain uncertain because most competitor reports do not provide a verified input resolution together with an explicit FLOP-counting convention. No matched-hardware latency benchmark was performed, so the present evidence does not establish real-time inference suitability.

Placement Analysis

The multi-scale refinement layer is placed at encoder Stage 3 in the final FAFEM + RFG-MSCB configuration. This placement was selected based on observed results from the ablation chain. A single-seed exploratory all-skip configuration used batch size 32 rather than 24 in the main protocol; the available training evidence did not improve relative to the Stage-3 configuration. However, this comparison is limited by incomplete training and the batch-size mismatch, so it does not support a general conclusion about optimal placement. The Stage-3 placement provides a favorable trade-off between performance and computational cost in our experiments.

Limitations

- **No statistical significance testing:** All reported differences are descriptive means over three seeds; no hypothesis testing was performed.
- **Single-seed placement evidence:** The exploratory all-skip placement run is single-seed, incomplete, and uses a different batch size; it cannot establish a general placement rule.
- **No disentanglement of RFG-MSCB components:** The ablation isolates the combined residual frequency-guided coupling; individual contributions of the residual formulation, initialization strength, and frequency conditioning are not separately measured.

- **Incomplete experiments:** The residual-guidance run with default initialization did not complete training; its potential findings are unavailable.
- **Domain gap on external datasets:** Performance drops substantially on CVC-ColonDB and ETIS-LaribPolypDB, indicating limited out-of-distribution generalization.
- **Hardware and software not fully verified:** GPU model, CUDA/cuDNN versions, and `py_sod_metrics` version are not formally recorded.
- **Training duration:** Seed 7777 stopped early at epoch 167 (patience 30), while seeds 42 and 6543 reached epoch 199; this asymmetry may affect the mean estimate.

Future Work

- Systematic multi-seed placement ablation with controlled batch size and learning rate.
- Disentanglement of the RFG-MSCB components (residual form, initialization schedule, frequency conditioning) via factorial experiments.
- Investigation of initialization sensitivity using completed default-initialization runs.
- Domain adaptation or targeted augmentation to improve generalization on CVC-ColonDB and ETIS-LaribPolypDB.
- Statistical significance testing with larger seed ensembles or bootstrap confidence intervals.
- Formal recording of hardware, software, and dependency versions for full reproducibility.

Conclusion

We presented a ConvNeXt-Tiny U-shaped segmentation framework that introduces a cross-level frequency-guided multi-scale refinement mechanism. The model integrates a bottleneck Frequency-Aware Feature Enhancement Module (FAFEM) with a Residual Frequency-Guided Multi-Scale Convolution Block (RFG-MSCB) placed at the Stage-3 encoder-decoder fusion. The frequency descriptor extracted by FAFEM is propagated to the decoder and used to condition the branch weights of the MSCB through a bounded residual formulation.

The RFG-MSCB preserves the conventional multi-scale pathway as a reference and learns a controlled, sample-dependent deviation guided by the frequency descriptor. This coupling connects deep semantic frequency information with higher-resolution spatial feature processing, while the remaining decoder stages are kept lightweight.

Ablation experiments using three random seeds show that the addition of the Stage-3 MSCB-lite to a FAFEM baseline increases Kvasir-SEG mDice by +0.34 percentage points. The further addition of the residual frequency-guided coupling yields a final Kvasir-SEG mDice of 0.9313 ± 0.0044 , an additional +0.19 pp over the MSCB-lite configuration. On external datasets, the model achieves mDice of 0.9370 on CVC-ClinicDB, 0.8821 on CVC-300, 0.7944 on CVC-ColonDB, and 0.7870 on ETIS-LaribPolypDB. Performance on CVC-ColonDB and ETIS remains lower, indicating limited out-of-distribution generalization.

The final architecture adds 2.4% in parameters relative to the FAFEM-only configuration; its detailed complexity and measurement convention are reported in Table 1. All performance values are descriptive means over three seeds; no statistical significance testing was performed.

In summary, this work demonstrates that cross-level frequency guidance with bounded residual branch adaptation is a viable design for polyp segmentation. Future work includes systematic placement ablation, component disentanglement of the RFG-MSCB, and domain adaptation strategies for stronger generalization on challenging external datasets.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

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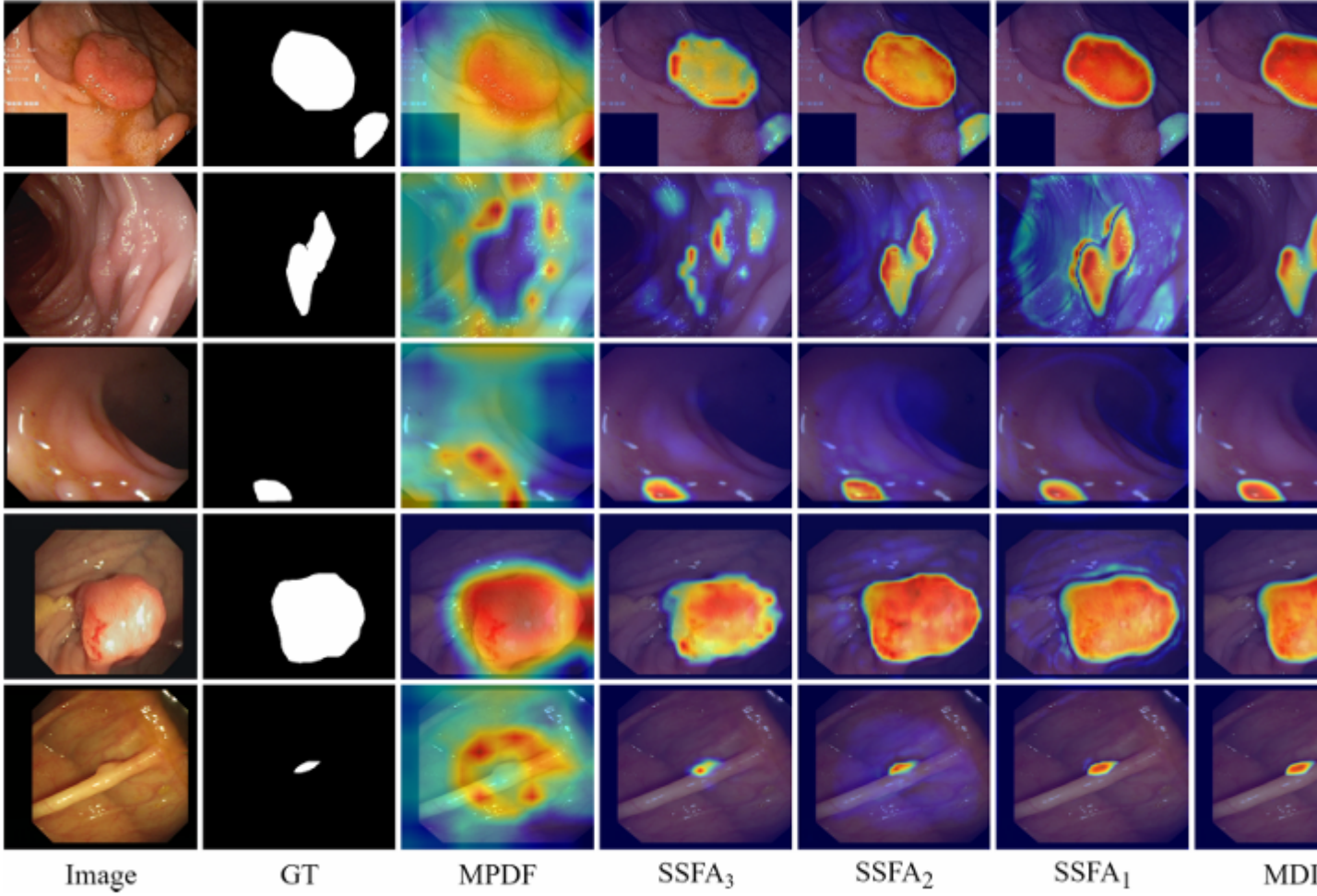


Fig. 6 Attention heatmaps for the model's MPDF, SSFA_i, and MDI modules. Darker colors indicate higher attention

- **MultiScaleContext**: tests whether multi-scale/global context helps.
- **BSEI**: tests boundary-aware feature enhancement.
- **DetailBranch**: tests high-resolution spatial detail recovery.
- **GatedDetailFusion**: tests whether adaptive fusion is better than simple fusion.

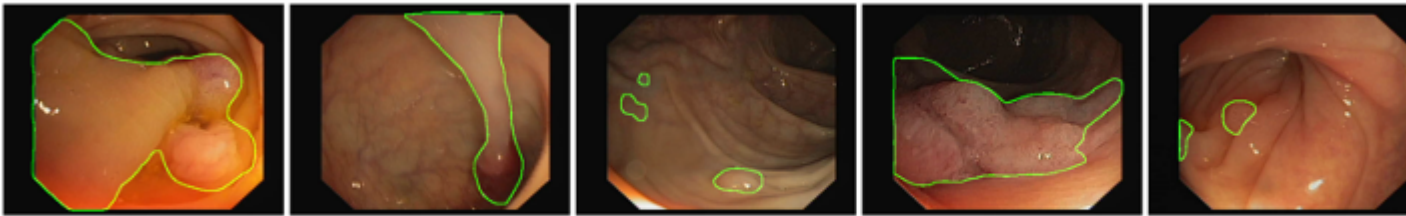


Fig. 1 Six examples of polyp images, with the circled green area in each image representing the polyp

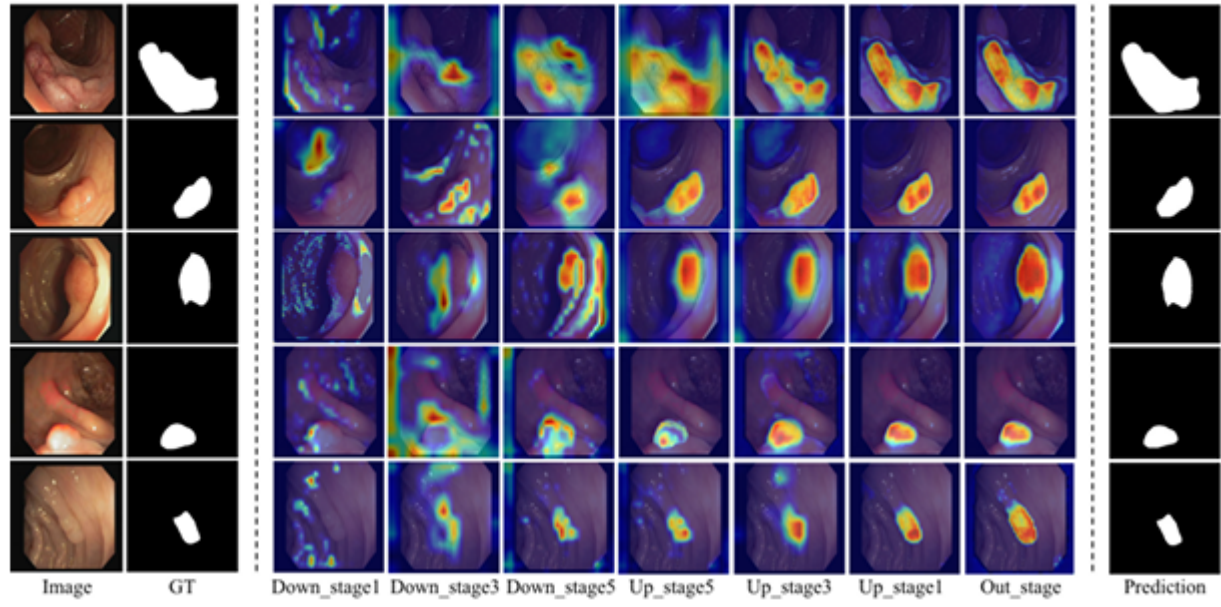


Fig. 8. Module visualization feature map.

TABLE VII
ABLATION STUDY ON BACKBONE WIDTH (BASE_CHANNEL) AND THE COMPUTATIONAL OVERHEAD OF HCA.

base_channel	HCA	CVC-ClinicDB		ETIS-LaribPolypDB		Param (MB)	GFLOPs (G)
		mDice	mIoU	mDice	mIoU		
7	–	0.801	0.748	0.662	0.627	3.45	2.15
7	+	0.902	0.847	0.792	0.742	5.12	3.39
10	–	0.822	0.773	0.693	0.642	6.82	4.05
10	+	0.928	0.879	0.812	0.763	8.94	6.01
12	–	0.843	0.797	0.721	0.672	7.92	6.21
12	+	0.953	0.909	0.833	0.790	10.13	9.33
14	–	0.846	0.802	0.724	0.676	10.77	8.43
14	+	0.956	0.911	0.837	0.794	13.68	12.09
17	–	0.851	0.805	0.729	0.681	15.91	12.11
17	+	0.958	0.913	0.840	0.799	20.36	17.98

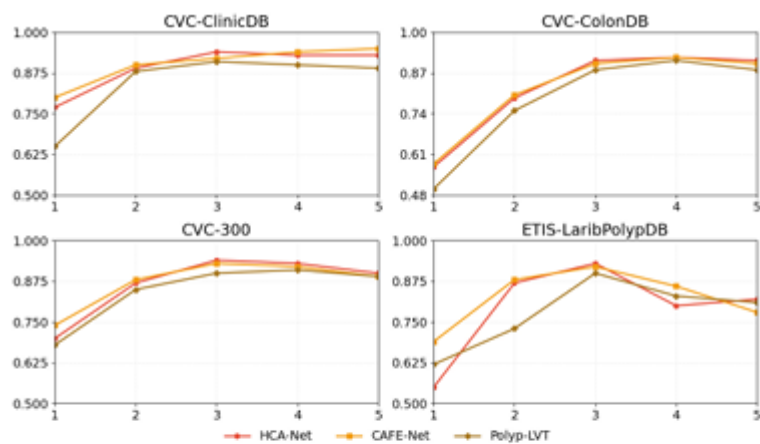


Fig. 6. Performance vs. size on the five polyp datasets. The x-axis denotes the proportion (%) of polyp size, and the y-axis represents the averaged mDice coefficient.

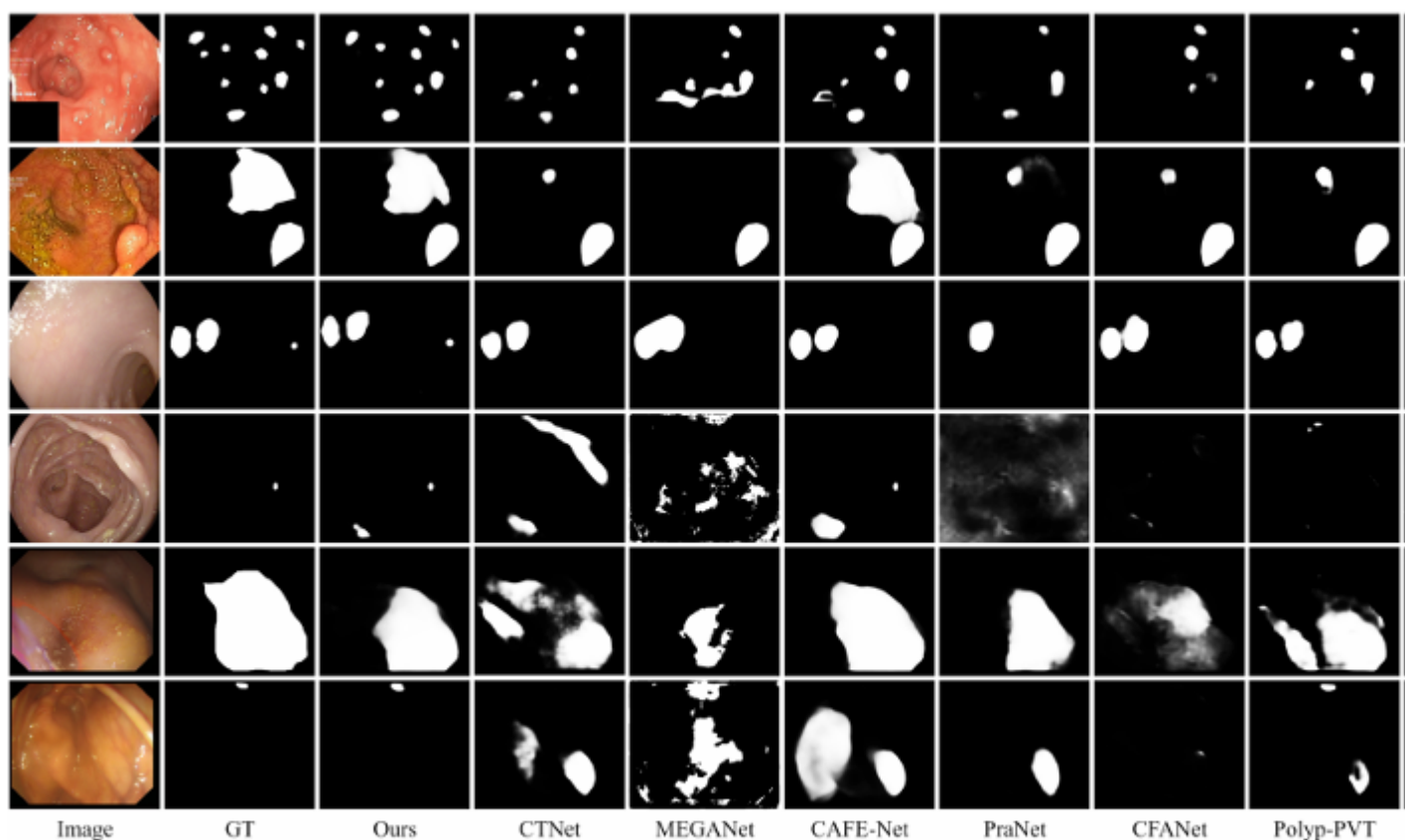


Fig. 4 Our model was compared with the results of various advanced methods on the polyp dataset

- Image(Diagram) of the model with related Modules -> two images
- Heatmap of the model in ablation study with 6 samples
- Heatmap of the model for some stages

- Heatmap of the model in comparison with other models
- Table of the ablation study